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ENGR-E583 INFORMATION VISUALIZATION MAPPING AND ANALYZING PARTICIPATION - TEAM K

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Network-Map (Open-Source Template) How-To Guide - Documentation

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1. Project Overview

NetworkMap is an event network visualization platform that helps event organizers **collect participation and connection data**, automatically convert it into **network-ready nodes and edges**, and explore insights through an **analytics dashboard**. Outputs can be exported to tools like **Kumu.io** and **Gephi** for deeper relationship mapping and network analysis.

Primary Use Case

This template supports projects like the Tutor/Mentor Institute's goal of creating an **open-source workflow** that transforms event registration data into network maps that reveal:

- Which organizations participated across time
- Where relationships and repeat participation exist
- Which organizations are missing (dropped off) across events
- How participation varies by sector/location

2. Live Links

- **Live Website (Netlify):** <https://network-mapping-template.netlify.app/>
- **Getting Started Page:** <https://network-mapping-template.netlify.app/learn/getting-started>
- **GitHub Repository:** <https://github.com/Ramcharxn/Mapping-Analyzing-Participation>

3. Key Features

For Event Organizers (Admins)

- Create accounts (Sign Up / Log In)
- Build event forms (with participant + connection fields)
- View analytics dashboards (demographics + connection metrics)
- Export data to **CSV** for Kumu/Gephi
- Convert uploaded event data into **nodes and edges**

For Participants

- Fill event registration forms
- Add up to **5 organizational connections**
- Use guided location selection (Country → State → City)

4. Supported Data Formats

The application supports the following data formats.

- CSV
- Excel (.xlsx)

5. Local Setup & Installation

Prerequisites

- Node.js v14+
- MongoDB (local or MongoDB Atlas)
- Git

Step A: Clone the Repository

```
git clone https://github.com/kishuxz/info_viz.git  
cd info_viz/event-network-template
```

Step B: Install Dependencies

Frontend

```
npm install
```

Backend

```
cd backend  
npm install  
cd ..
```

Step C: Configure Environment Variables

Create .env files by copying the examples:

- backend/.env from backend/.env.example
- /.env from .env.example in the root

Backend: backend/.env

MONGODB_URI=your_mongodb_connection_string

JWT_SECRET=your_secure_random_string

NODE_ENV=development

PORT=5001

Frontend: .env (project root)

REACT_APP_API_URL=http://localhost:5001/api

Step D — Run the Application (Two Terminals)

Terminal 1 (Backend)

cd backend

npm start

Terminal 2 (Frontend)

npm start

Default Local URLs

- Frontend: http://localhost:3000
- Backend: http://localhost:5001

6. Application Navigation

Home Page

The home page explains the platform value and provides primary calls to action like:

- **Start Mapping**
- **View on GitHub**
- Navigation: Home, Learn, Features, GitHub, Log In, Sign Up

Learn Section

Contains educational modules and step-by-step help for:

- Understanding network mapping concepts
- How the workflow goes from **forms/spreadsheets** → **nodes/edges** → **Kumu/Gephi**

Split / Convert Page (Upload & Convert Event Data)

This is where you upload CSV/Excel and convert into a network format.

Step 1: Choose Graph Type

Common modes include:

- **Organizations ↔ Events** (bipartite attendance mapping)

- **Organizations ↔ Organizations** (direct relationships between orgs)
- **Custom A → B** (choose any two columns as source/target)
- **Sector-based (beta)** options like Event → Sector → Organization (when sector exists)

Step 2: Upload Files

- Click **Choose Files**
- Select your CSV/Excel
- Choose an output format (e.g., **Gephi (Source/Target)**)

Step 3: Convert

- Click **Convert to Network**
- Download the generated outputs (nodes + edges)

7. How to Use the Platform (End-to-End Workflow)

Workflow Option 1: Organizer Form → Analytics → Export

1. **Sign Up / Log In** as an Admin
2. Go to **Admin Dashboard** → **Create Form**
3. Enter event metadata (event name/date)
4. Add/confirm fields (participant details + connection fields)
5. Save and generate a **shareable form link**
6. Share the link with participants
7. After responses are collected, open **Analytics Dashboard**
8. Use **Export** to download CSV outputs for mapping tools

Workflow Option 2: Spreadsheet Upload → Convert → Kumu/Gephi

1. Open the **Split/Convert** page (/split)
2. Choose the appropriate **Graph Type** (most common: **Organizations ↔ Events**)
3. Upload CSV/Excel
4. Click **Convert to Network**
5. Download generated nodes/edges (or Source/Target edges)
6. Import into **Kumu** or **Gephi**

8. Dashboards: How to Open & Navigate

Opening Dashboards

- Log in as Admin
- Navigate to **Analytics** (or “View Analytics Dashboard” button if available on landing sections)

What You will See

Dashboards are designed to answer “Who participated?” and “How are connections forming?” using:

- Summary metrics (e.g., number of connections, participants, etc.)
- Charts for participation patterns
- Connection statistics (how many links people report, common orgs, etc.)

Filters & Interactions (Typical)

Depending on your dataset and enabled UI:

- Filter by **Event**
- Filter by **Date range**
- Filter by **Location** (Country/State/City)
- Filter by **Organization / Sector**
- Hover tooltips for exact values
- Click interactions to focus/highlight categories

Tip: If you are analyzing “who missed out”, use the dashboard and exports together:

- Dashboard shows overall participation patterns
- Exported nodes/edges in Kumu/Gephi makes “drop-offs” visually obvious (orgs that appear in early events but not later ones)

9. Exporting to Kumu and Gephi

Export Formats

NetworkMap supports exporting processed data as CSV files so you can use:

- **Kumu.io** (relationship mapping + storytelling views)
- **Gephi** (metrics + community detection + layouts)

Import into Kumu (Recommended Steps)

1. In Kumu, create a new project
2. Import **Elements (Nodes)** CSV first
3. Import **Connections (Edges)** CSV next
4. Confirm that:
 - Node “label”/“name” fields map correctly
 - Edge “from” and “to” (source/target) fields map correctly
5. Use Kumu styles:
 - Color by type (Event / Organization / Sector)
 - Size nodes by degree (most connected)
 - Filter by event/date to see change over time

Import into Gephi (Recommended Steps)

1. Open Gephi → New Project
2. Import edges as **Source/Target**
3. If nodes are separate, import nodes table
4. Run:
 - Layout (e.g., ForceAtlas2)
 - Statistics (degree, modularity)
5. Use ranking to size nodes by centrality

10. Security & Data Handling Notes

- Passwords are hashed with **bcrypt**
- Inputs include validation (emails, phone numbers, required fields)
- CORS enabled and configured for web access
- You can self-host for full control of data storage

11. Troubleshooting (Common Issues)

Backend won't connect to DB

- Verify MONGODB_URI is correct

- If using Atlas, ensure IP whitelist allows access (often 0.0.0.0/0 for testing)

Frontend could not reach backend

- Confirm REACT_APP_API_URL matches backend URL
- Check backend is running on PORT=5001

Render “cold start” delay

- Free tiers may spin down after inactivity; first request may take longer

Large CSV uploads

- Very large networks may take time to parse and render; try smaller subsets first

12. Deployment Summary (Production)

Deploy Backend (Render)

- Root Directory: backend
- Build: npm install
- Start: npm start
- Set env vars: MONGODB_URI, JWT_SECRET, NODE_ENV=production, PORT=5001

Deploy Frontend (Netlify)

- Build: npm run build
- Publish: build
- Set env var:

REACT_APP_API_URL=https://<your-render-backend>/api

13. License & Attribution

- License: **MIT**
- Client project context: Tutor/Mentor Institute work should include appropriate recognition per sponsor request (crediting the source of the idea and linking to sponsor site where required).